

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY

IN-CLASS MIDTERM, 2:40-3:55 ON 4/1/26

6 numbered problems

Closed-book; no notes, computers, calculators, nor visible phones

Unless explicitly stated otherwise, you may use any result covered in class. If your solution makes use of a theorem from class, make sure to explain why it applies (e.g., if the theorem has hypotheses).

Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)

It is not sufficient to state that a problem is identical to one from a previous homework or practice exam, when that is the case.

If you run out of space, use the back side of the paper and indicate this clearly.

Name: _____

Uni: _____

1. (5 points for each lettered question.)

Let A , B , C , and D be abelian groups, and $f : A \rightarrow B$, $g : B \rightarrow C$, and $h : C \rightarrow D$ be homomorphisms.

- (a) State the definition of what it means for the sequence

$$A \xrightarrow{f} B \xrightarrow{g} C$$

to be exact.

- (b) Prove that if

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$$

is exact and f is surjective, then h is injective.

- (c) Give an example of a short exact sequence of the form

$$0 \rightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{f} \mathbb{Z}/4\mathbb{Z} \xrightarrow{g} \mathbb{Z}/2\mathbb{Z} \rightarrow 0.$$

Your answer should define f and g . In this problem, you do not need to justify exactness.

- (d) Let (A_*, ∂^A) and (B_*, ∂^B) be chain complexes. State what it means for $f : (A_*, \partial^A) \rightarrow (B_*, \partial^B)$ to be a chain map. Then explain how the induced map $H_p(f) : H_p(A_*, \partial^A) \rightarrow H_p(B_*, \partial^B)$ is defined.

For full credit, you should write a formula for $H_p(f)$ and briefly explain why it is well defined.

2. (5 points for each lettered question.)

Let C be a category, and let c , c' , and c'' be objects of C .

(a) Let $f \in C(c, c')$ a morphism. State the definition of what it means for f to be an isomorphism.

(b) Prove that if $F : C \rightarrow D$ is a functor and $f \in C(c, c')$ is an isomorphism, then $F(f)$ is also an isomorphism.

In this problem, it is not sufficient to state "this was proved in class" or similar.

(c) Prove that if $f \in C(c, c')$ is an isomorphism, then the function

$$C(c', c'') \rightarrow C(c, c'')$$

$$f' \mapsto f' \circ f$$

is a bijection.

3. (5 points for each lettered question.)

(a) State the excision theorem.

(b) Let M be a Hausdorff space, $U \subset M$ an open subset, and assume that U is homeomorphic to $\mathbb{R}^n$. Prove that

$$H_p(M, M \setminus \{x\}) \cong H_p(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$$

for all $x \in U$ and all p .

(c) State the homotopy invariance theorem.

4. (5 points for each lettered question.)

(a) State a formula for the homology group $H_p(S^n)$.

For full credit, you must state a correct formula for all p and n . In this problem, you do not need to justify your answer.

(b) Let $S \subset \mathbb{R}^5$ be a subspace, and assume that S is homeomorphic to S^2 . Determine a formula for $H_p(\mathbb{R}^5, S)$ for all p .

Justify your answer.

5. (5 points for each lettered question.)

(a) State a formula for $H_p(D^{n+1}, S^n)$ for all p .

For full credit, you must state a correct formula for all p and n . In this problem, you do not need to justify your answer.

Now let $n \geq 1$ and let $f : S^n \rightarrow S^n$ be a continuous map.

(b) State the definition of *degree* of f .

(c) Let $F : D^{n+1} \rightarrow D^{n+1}$ be defined as

$$F(x) = \begin{cases} |x|f(\frac{x}{|x|}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

It may be used without proof that F is continuous. Prove that the induced homomorphism

$$H_{n+1}(F) : H_{n+1}(D^{n+1}, S^n) \rightarrow H_{n+1}(D^{n+1}, S^n)$$

is given by $x \mapsto dx$, where d is the degree of f .

6. (2 points for each lettered question.)

For each of the following statements, mark if they are true or false. *In this problem, you do not need to justify your answers.*

- (a) If $F : \mathbf{Sets} \rightarrow \mathbf{Sets}$ is a functor, then $F(\emptyset) = \emptyset$.
- (b) If X and Y are sets, and $f : X \rightarrow Y$ is a bijection, then $F(f)$ is a bijection for any functor $F : \mathbf{Sets} \rightarrow \mathbf{Sets}$.
- (c) If X is simply connected, then $H_1(X) \cong 0$.
- (d) If X and Y are homotopy equivalent, then $H_p(f) : H_p(X) \rightarrow H_p(Y)$ is an isomorphism for any continuous map $f : X \rightarrow Y$ and any p .
- (e) If $c : X \rightarrow Y$ is a constant map, then $H_p(c) : H_p(X) \rightarrow H_p(Y)$ is zero for all $p \geq 1$.
- (f) If $x \in X \subset \mathbb{R}^n$, then $H_p(\mathbb{R}^n, X) \cong H_{p-1}(X, \{x\})$ for all p .
- (g) If $S^n \simeq S^m$ then $n = m$.
- (h) For any abelian groups A and B , there exists a short exact sequence of the form

$$0 \rightarrow A \rightarrow A \oplus B \rightarrow B \rightarrow 0.$$